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Comparison of the Structures of the Quinone-binding Sites in Beef Heart Mitochondria*

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The ubiquinone pool in mitochondrial membranes serves as an electron carrier between both NADH-coenzyme Q oxidoreductase (Complex I) and succinate-coenzyme Q oxidoreductase (Complex II) and ubiquinol-cytochrome-*c* oxidoreductase (Complex III). It has been reported (Saitoh, I., Miyoshi, H., Shimizu, R., and Iwamura, H. (1992) *Eur. J. Biochem.* 209, 73–79) that 2-alkyl-4,6-dinitrophenols compete with exogenous coenzyme Q (Q) to inhibit electron transport through cytochromes *b* and c_1 in mammalian mitochondria as well as in photosystem II. We have probed the similarities and differences in the reaction sites of exogenous Q in all three segments of the respiratory chain using selected 2-alkyl-4,6-dinitrophenols. The inhibition of Q analog reduction by the dinitrophenol derivatives was competitive for Complex I and noncompetitive for Complex II. The inhibition of Complex III was competitive with the pentyl analog, but was uncompetitive with the decyl analog, which may be due to different interactions of the two quinol analogs with Complex III. The degree of inhibition by several of these compounds was comparable for Complexes I and III, but Complex II was inhibited to a much smaller extent. The inhibitory potency of these compounds for Complexes I and III was increased by branching and by lengthening the carbon chain at the 2-position equivalent to the isoprenoid side chain of ubiquinone. Hydrophobic substituents increased the inhibition of Complex II. Replacement of the phenolic OH group by a chlorine atom decreased the maximum inhibition of Complex III, but increased that of Complex I. These data suggest that the structures of the exogenous Q-binding sites in Complexes I and III may be similar, but not identical, and that they are different from that in Complex II.

The quinones in energy-transducing biomembranes act as mobile carriers of electrons and protons and play a central role in transferring reducing equivalents between several dehydrogenases and the cytochrome bc_1 segment of the respiratory chain in mammalian mitochondria. Current knowledge of the structures of the Q[1]-binding sites in Complexes I, II, and III is based largely on the differential effects of inhibitors and substrate analogs. Quinone analog inhibitors such as heptylhydroxyquinoline *N*-oxide seem to act selectively in the cytochrome bc_1 segment, with little or no effect on Complexes I and II (1, 2), suggesting structural differences between the Q-binding sites in the two dehydrogenases and those in the cytochrome complex. The Q cycle mechanism proposed for mammalian Complex III requires two Q interaction sites (1, 3). One of these, the Q_o site, is sensitive to myxothiazol, but not to antimycin A, whereas the other, the Q_i site, is inhibited by antimycin A, but not by myxothiazol (1, 3). Furthermore, the various mutations leading to inhibitor resistance map on opposite membrane sides of cytochrome *b*, suggesting that the Q_o and Q_i sites are separate and different (4).

Based on the observations of interacting Q radical signals after reduction, Complex II is also thought to have two interacting Q sites (5). Thenoyltrifluoroacetone and carboxins, specific inhibitors of Complex II, give hyperbolic inhibition curves, suggesting a single binding site for these inhibitors, in agreement with data for the binding of labeled carboxins (6, 7). Interactions with Q analog acceptors indicate only one type of site of interaction with exogenous Q (8).

Less is known about the Q-binding site(s) in Complex I, but Q radical signals have been observed in coupled particles. The "classical" inhibitors rotenone and piericidin A and barbiturates all act at the same point between the high potential iron-sulfur cluster and Q (9). Since these inhibitors compete for the binding sites (2 eq of piericidin are required for full inhibition), they are very likely to be bound at or very near the same site in the protein (10, 11). The same is true of MPP^+ and especially of its 4′-alkyl-substituted analogs, which prevent the binding of and inhibition by rotenone and piericidin A (12, 13). Full inhibition by MPP^+ analogs also involves two differing sites of inhibition, one more hydrophobic than the other (14). The structural features of the rotenone site(s) have also been probed using 4-hydroxyquinoline derivatives as inhibitors (15). A recent report (16) that rotenone and Q analog substrates do not bind at exactly the same point in Complex I suggests that the inhibitors must prevent the passage of electrons from the high potential Fe-S cluster of the enzyme to Q rather than competing with Q. Therefore, structure-function studies with inhibitors acting at the rotenone site(s) do not provide information about the Q-binding site in Complex I.

In view of the dearth of knowledge of the structures and locations of the Q-binding sites in the three complexes, it would

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[1] The abbreviations used are: Q, coenzyme Q; Complexes I, II, III, and IV, NADH-coenzyme Q oxidoreductase, succinate-coenzyme Q oxidoreductase, cytochrome bc_1 (ubiquinol-cytochrome-*c* oxidoreductase), and cytochrome-*c* oxidase segments, respectively, of the intact electron transport chain; ETP, electron transport particles (an inverted submitochondrial particle preparation); DB, 2,3-dimethoxy-5-methyl-6-decyl-1,4-benzoquinone; DPB, 2,3-dimethoxy-5-methyl-6-pentyl-1,4-benzoquinone; DBH_2 and $DPBH_2$, reduced (quinol) forms of DB and DPB, respectively; MPP^+, 1-methyl-4-phenylpyridinium ion; TPB^-, tetraphenylboron ion; I_H, concentration required to give half of the maximum inhibitory effect.

be useful to develop a series of Q analogs that inhibit Complexes I, II, and III to define similarities and differences among the Q sites in the relative degrees of inhibition by a given compound. The 2-substituted 4,6-dinitrophenols previously studied by Saitoh *et al.* (17) as inhibitors of Complex III and of the Q_B site in photosystem II seemed suitable. The conformation of the alkyl substituents at the 2-position, corresponding to the isoprenoid side chain of ubiquinone, was the significant factor for inhibitor binding to the ubiquinone-binding site, irrespective of substituent patterns at the 4- and 6-positions (18). The stereochemical similarity between the 2-substituents of potent inhibitors and the side chain of ubiquinone was demonstrated in conformational studies by molecular orbital calculations (17).

We have now extended these studies on the inhibitory effects of 2-substituted 4,6-dinitrophenols to Complexes I and II and compared the results with their effect on Complex III using inverted submitochondrial particles (electron transport particles (ETP)) from beef heart. Although quinones of varied structure have been used to explore some of the Q-binding sites (*e.g.* Refs. 8 and 19), to our knowledge, this is the first systematic study comparing structure-inhibition patterns in all three Q-dependent segments of the respiratory chain with the same inhibitors.

EXPERIMENTAL PROCEDURES

Materials

Inverted mitochondrial inner membranes were prepared from beef heart mitochondria according to the method of Crane *et al.* (20). 2-Substituted 4,6-dinitrophenols were synthesized, and fresh aqueous solutions were prepared just before use as described (17). The hydrophobicity index (π value) for each analog listed in Table II was estimated from the partition coefficient (octanol/water). For compounds 6, 8–10, 15, 16, and 23, fragmental factors of 0.54 and −0.13 were adopted for $\pi(CH_3)$ and the branching factor (F_{cBr}), respectively (21). The π values of the substituents of compounds 1–4 and 7 were cited from the literature (22). DB was purchased from Sigma. DPB was the kind gift of Dr. G. White. The reduced quinone analogs DBH_2 and $DPBH_2$ were prepared by adding borohydride to an anaerobic solution of the analog in ethanol, 50 mM Na^+ borate, pH 7.2 (9:1, v/v). The product was extracted into anaerobic cyclohexane, dried, and then stored in acidified ethanol under argon.

Assays

All values presented were averaged from at least two determinations, and the standard deviation was within 5%. Kinetic patterns were identified from computer-fitted plots of the data using Enzfitter (Biosoft, Cambridge, United Kingdom), but the plots in Figs. 1–3 were drawn manually for illustrative purposes. Each experiment was done with two different inhibitor analogs to ensure that the kinetic patterns were not specific for a certain analog.

Complex I—NADH-Q oxidoreductase activity was measured spectrophotometrically by the decrease in the absorbance due to NADH at 340 nm. The assay cuvette contained 0.25 M sucrose, 50 mM phosphate, pH 7.6 (at 30 °C), 1.0 mM potassium cyanide, 0.28 mM NADH, 60 μM DB or DPB, and 20 μg/ml ETP.

Complex II—The oxidation of succinate by DB or DPB was coupled to the reduction of dichloroindophenol, and the rate was followed spectrophotometrically at 600 nm at 30 °C. The reason for coupling the oxidation with dichloroindophenol instead of monitoring the reduction of the quinone analog directly is that the absorbance of the dinitrophenols at 276 nm interferes with the assay. The assay mixture contained 0.25 M sucrose, 50 mM phosphate, pH 7.6, 60 μM DB or DPB, 1.0 mM potassium cyanide, 0.05 mM dichloroindophenol, 1.0 mM EDTA, 20 mM succinate, and 20 μg/ml ETP. ETP (30 mg/ml) were preincubated at 37 °C with 1 mM malonate for 10 min to dissociate tightly bound oxalacetate and thus to fully activate the succinate dehydrogenase (23). The amount of malonate carried over into the assay was negligible.

Complex III—Succinoxidase activity, after activation as described above, was assayed polarographically at 30 °C in 0.25 M sucrose, 50 mM phosphate buffer, pH 7.6. Inhibition of Complex II was negligible at the concentrations of inhibitor used. The rate of electron flow through Complex III was also measured as reduced Q analog oxidase or as reduced Q analog–cytochrome *c* reductase in the same buffer with reduced DBH_2 or $DPBH_2$ (10 μM) and 23 μM cytochrome *c*.

Complex IV—Cytochrome oxidase activity was determined polarographically at 30 °C in 0.25 M sucrose, 50 mM phosphate, pH 7.6, after addition of 5 μg/ml reduced cytochrome *c*. Cytochrome *c* (10 mg/ml) was reduced with sodium dithionite and monitored spectrophotometrically at 550 nm. No inhibition was observed.

RESULTS

Site of Inhibition of 2-Substituted 4,6-Dinitrophenols in Complexes I and II—To interpret the kinetic data obtained with the inhibitors, it was important to establish that the inhibition was fully reversible, as expected for equilibrium interaction with a binding site, and also to identify where these dinitrophenols act in relation to known inhibitors. The reversibility was established for NADH oxidase by incubating a concentrated suspension of ETP (3 mg/ml) with compound 6 or 16 (see Table II for structures) for 5 min and assaying after 100-fold dilution into the assay medium with or without inhibitor at the same concentration as in the preincubation. Without inhibitor in the assay medium, the activity was the same as in the untreated control, despite maximum inhibition in the presence of inhibitor (data not shown). Dilution was sufficient to reverse the inhibition, as expected for a reversible inhibitor.

It has recently been suggested that rotenone and the Q analog DB bind to isolated Complex I at different sites (16). However, isolated Complex I lacks one of the two rotenone-binding sites present in more intact preparations. ETP have two slightly different binding sites for rotenone, both of which must be occupied for complete inhibition (11). Rotenone, piericidin A, and MPP^+ bind to these sites in a mutually exclusive manner (13). MPP^+ alone inhibits <50% of the NADH oxidase activity in ETP; but in the presence of the ion-pairing agent tetraphenylboron, complete inhibition is observed, again supporting the concept of two nonidentical inhibitory sites (11, 13), one of which has a hydrophobic barrier to the access of the hydrophilic MPP^+ molecule. When the kinetics of the inhibition of NADH-Q oxidoreductase were studied using DPB as the quinone acceptor, mixed inhibition by 4′-pentyl-MPP^+ plus 10 μM TPB^- (Fig. 1*A*) was observed. When we examined the kinetics of the inhibition of NADH-Q oxidoreductase by compound 16 using DB as the quinone acceptor, the pattern of inhibition was competitive (Fig. 1*B*). (TPB^- has no effect on the inhibition by compound 16, which is not positively charged.) The same results were obtained using DPB as the oxidant (data not shown). Thus, unlike the MPP^+ analogs, which compete with rotenone, the 4,6-dinitrophenol inhibitors compete directly with Q analog substrates.

In contrast, the kinetics of the oxidation of succinate with Q analogs (DB or DPB) as electron acceptors gave a clearly noncompetitive pattern, in which the K_m for the externally added Q analogs was unaffected by the 2-alkyl-4,6-dinitrophenol inhibitors (Fig. 2). This suggests that in Complex II, these inhibitors and the Q analogs do not act at the same site.

Inhibition of Complex III by 2-Substituted 4,6-Dinitrophenols—In the studies of Complexes I and II, the patterns of inhibition observed were identical whether DB or DPB served as the electron acceptor. Quite unexpectedly, with Complex III, the kinetic pattern observed with compound 8 as the inhibitor was competitive when reduced $DPBH_2$ served as the electron donor to cytochrome *c* (Fig. 3*A*), but was almost uncompetitive (the slopes increased slightly) when the reduced DBH_2/cytochrome *c* reaction was assayed (Fig. 3*B*). In both cases, the secondary plots (Fig. 3, *insets*) were parabolic, suggesting that more than one molecule of the inhibitor can bind. It should be noted that the maximum velocities obtained with the two electron donors are not the same. With $DPBH_2$, the V_{max} is 0.5 μmol of cytochrome *c* reduced per min/mg of protein, but with DBH_2, it is 1.7 μmol/min/mg. The more hydrophobic DBH_2 may penetrate

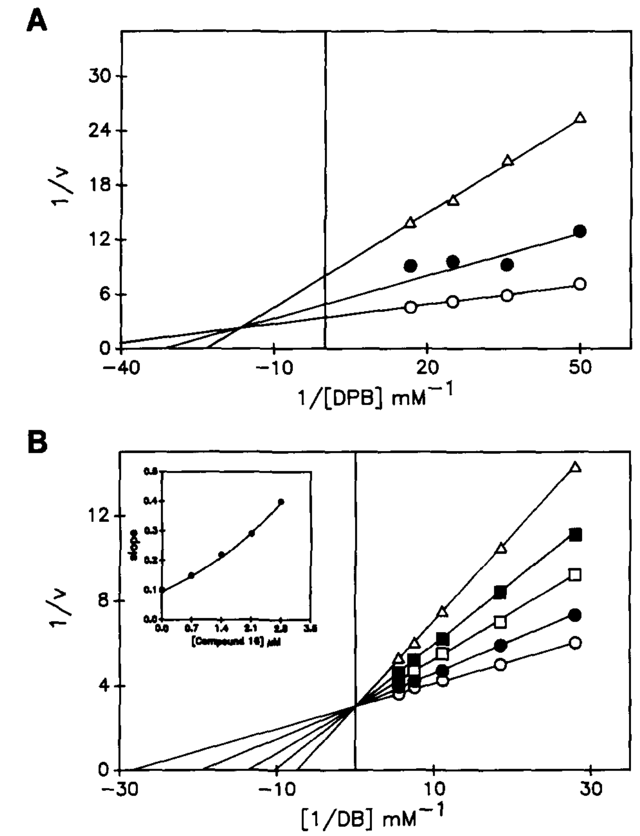


FIG. 1. **Double-reciprocal plots for inhibition of NADH oxidoreductase.** *A*, inhibition by 5′-pentyl-MPP^+ in the presence of 10 µM TPB^- using DPB as the oxidant. The inhibitor concentrations were 0 (○), 10 (●), and 20 (△) µM. *B*, inhibition by the 4,6-dinitrophenol analog compound 16 using DB as the oxidant. The inhibitor concentrations were 0 (○), 0.7 (●), 1.4 (□), 2.1 (■), and 2.8 (△) µM.

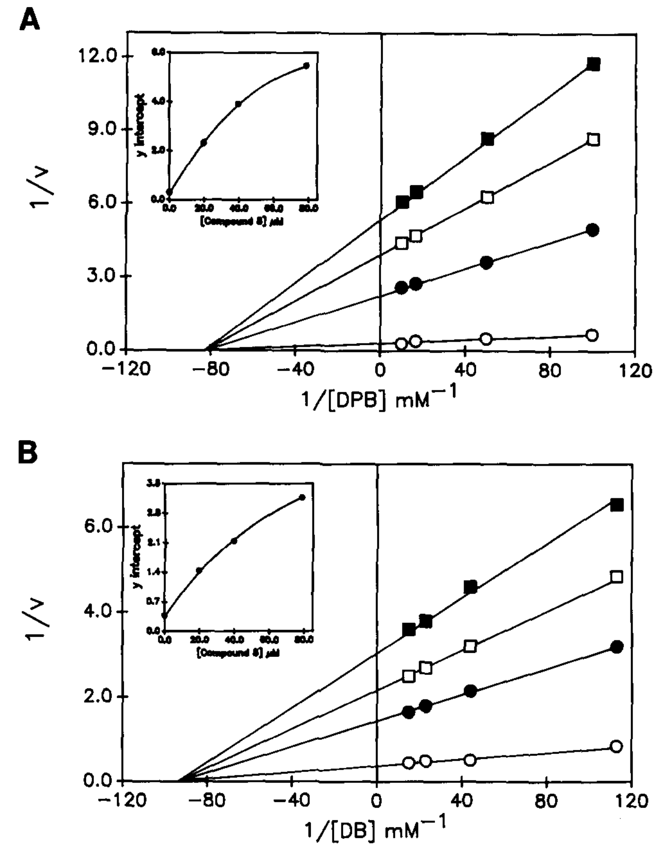


FIG. 2. **Double-reciprocal plots for inhibition of succinate-Q oxidoreductase.** *A*, noncompetitive inhibition by compound 8 using DPB as the oxidant; *B*, noncompetitive inhibition by compound 8 using DB as the oxidant. The inhibitor concentrations were 0 (○), 19.7 (●), 39.4 (□), and 78.7 (■) µM. The secondary plots (*insets*) show downward curvature as expected for partial inhibition.

deeper into the complex in the membrane to a site that facilitates more efficient electron transfer either to endogenous Q or to some component of the complex.

This means that the interactions of DBH_2 and $DPBH_2$ with Complex III must differ. To test whether a similar change in pattern occurs with inhibitors known to act at either Q_o or Q_i, we determined the effect of antimycin A and myxothiazol on the activity of Complex III with each Q analog. Fig. 4 shows the patterns obtained with $DPBH_2$. Myxothiazol, which acts at the Q_o site, inhibits noncompetitively (Fig. 4*A*), as expected for an inhibitor binding very strongly to the enzyme. Antimycin A, which acts at the Q_i site, inhibits uncompetitively (Fig. 4*B*), suggesting that the inhibitor acts in a different segment of the reaction mechanism from the electron donor. The same pattern for each inhibitor was obtained when DBH_2 was the substrate (data not shown). It is unlikely that the two quinol analogs interact at different points in Complex III. Instead, a minor difference in the positions of $DPBH_2$ and DBH_2 in a larger site may be responsible for the different inhibitory patterns obtained with the 4,6-dinitrophenol inhibitors. An analogous shift in the exact point of interaction was seen for the binding of the 4′-acyl-MPP^+ analogs to Complex I in that 4′-pentyl-MPP^+ and the shorter chain analogs prevented the binding of piericidin, but 4′-heptyl-MPP^+ (only two carbons longer) did not (13).

The inhibition patterns for all the inhibitors with each Q analog are given in Table I. A hexa-uni ping-pong mechanism with two irreversible steps has been proposed for Complex III (24). The simplest kinetic scheme is shown in Scheme 1*A*. The patterns obtained for myxothiazol and antimycin A are consistent with this scheme. To explain why the pattern of inhibition for a given inhibitor (the 4,6-dinitrophenol derivative) is different for two very similar quinol analog substrates, a more elaborate description of the interaction of the quinol with the enzyme is required. The parabolic inhibition (Fig. 3, *insets*) is consistent with the binding of at least two inhibitor molecules, so we propose a model for the substrate site in Complex III that has two substrate/inhibitor-binding sites (Scheme 1*B*). $DPBH_2$ competes directly at only the site closer to the surface, whereas DBH_2 interacts at either site. The possible inhibitor and substrate combinations shown in Scheme 1*B* could yield the observed kinetic patterns. The K_i values estimated for compound 8 from the secondary plots (Fig. 3, *insets*) were the same (0.04 µM) with either $DPBH_2$ or DBH_2 as the substrate, which is consistent with a scheme in which the inhibitor and substrates act at common sites.

Structure-Function Study of 2-Substituted 4,6-Dinitrophenols as Inhibitors of Respiratory Chain—The titration curves for the inhibition of each of the four respiratory complexes by compounds 6 and 16 are shown in Fig. 5. Complex IV is not inhibited at all by either compound, and Complex II is inhibited only at higher concentrations. Complexes I and III are both strongly inhibited at very similar concentrations of the inhibitors. Table II summarizes the sensitivities of the various complexes to inhibition by the nitrophenols. The potency of these phenols as inhibitors can be assessed in two ways: 1) the maximum percent inhibition attained and 2) the concentration that gives half of the maximum effect (I_H). I_H is used in preference to IC_{50} because, with many of the compounds, the inhibition leveled off at 50–65%. Where the maximum inhibition is 80% or higher, it does not matter which definition is used. However, where the inhibition is between 50 and 65%, a 2–3-fold differ-

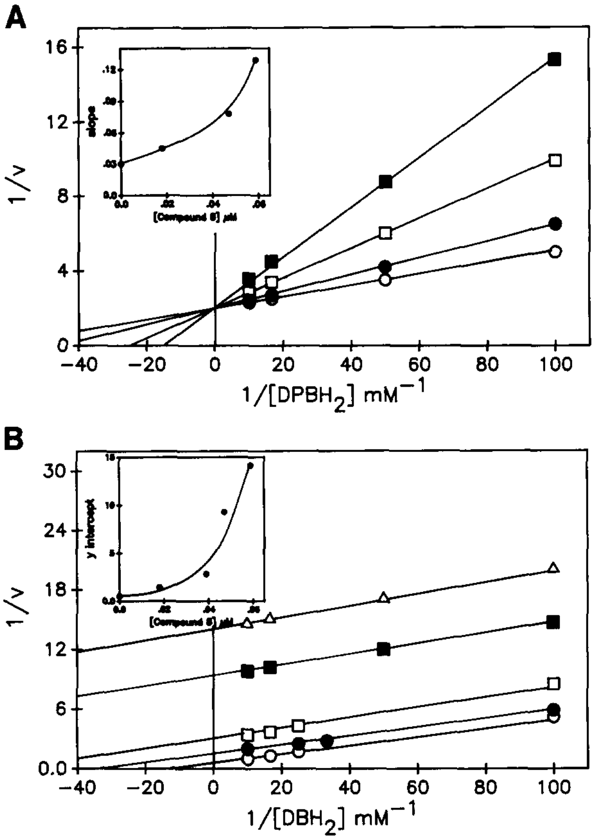


FIG. 3. **Double-reciprocal plots for inhibition of Complex III by 4,6-dinitrophenol analogs.** *A*, competitive inhibition by compound 8 of $DPBH_2$-cytochrome-*c* oxidoreductase. The inhibitor concentrations were 0 (○), 0.018 (●), 0.047 (□), and 0.059 (■) μM. *B*, uncompetitive inhibition by compound 8 of DBH_2-cytochrome-*c* oxidoreductase. The inhibitor concentrations were 0 (○), 0.018 (●), 0.039 (□), 0.047 (■), and 0.059 (△) μM. The secondary plots of the data (*insets*) show parabolic inhibition suggesting multiple inhibitor-binding sites.

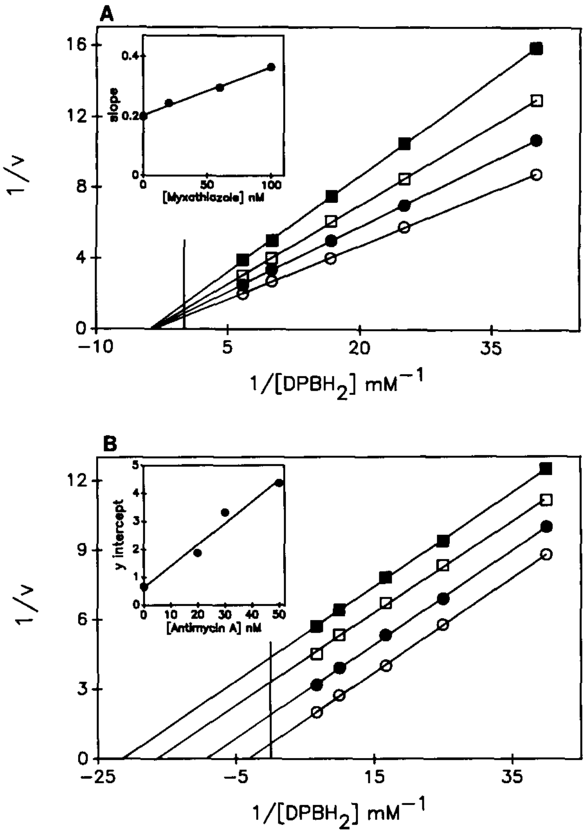


FIG. 4. **Double-reciprocal plots for inhibition of Complex III by specific inhibitors.** *A*, noncompetitive inhibition by myxothiazol, which acts at the Q_o site. The concentrations of myxothiazol were 0 (○), 20 (●), 60 (□), and 100 (■) nM. *B*, uncompetitive inhibition by antimycin A, which acts at the Q_i site. The concentrations of antimycin A were 0 (○), 20 (●), 30 (□), and 50 (■) nM. ETP were added last and $DPBH_2$ oxidase activity measured.

ence between I_H and IC_{50} values exists. To compare the efficacy of these partial inhibitors, we used the I_H to examine the structure-inhibitory activity relationship.

Inhibition of NADH-Q Oxidoreductase—The I_H values for the inhibition by the nitrophenols (Table II) were assessed in terms of the number of carbon atoms in the longest continuous chain of the 2-substituent. For the cyclohexyl derivative (compound 23), the carbon number was taken to be four. 4,6-Dinitrophenol (compound 1) was the poorest inhibitor. The efficacy tended to increase as the main chain length of the 2-substituent increased, with maximum potency at four carbons. The inhibition produced by derivatives with an α-branching structure was significantly higher than the inhibition produced by those lacking such a structure. This structure-activity relationship was similar to that previously reported for Complex III in rat liver mitochondria (17).

The potency in terms of the maximum percent inhibition was also considered. Incomplete inhibition is presumed to mean that the inhibition site is saturated with the inhibitor, but its presence does not completely block the passage of electrons to the Q analog acceptor. With Complex I, we found that the maximum inhibition levels for both compounds 9 and 10 were the same (63%), yet the I_H for compound 9 is 0.19 μM, whereas that for compound 10 is 85 μM. Compounds 1 and 2 inhibited slightly more of the activity (>80%), but the I_H values were >100 μM. These results indicate that the binding affinity is not directly related to the intrinsic inhibition of the catalytic reaction in ubiquinone redox sites.

The maximum inhibition is highest with compound 6A (90%),

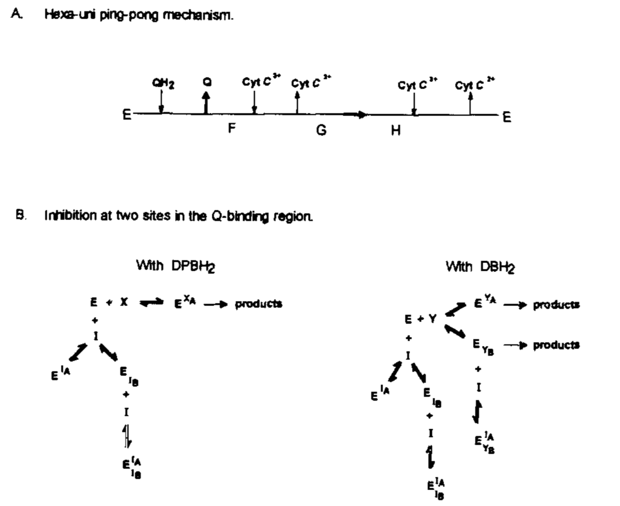
TABLE I
Inhibition patterns with respect to the Q analog substrate for Complexes I, II, and III

Complex I was assayed as NADH-analog reductase in the presence of cyanide. Complex II was assayed as succinate-Q analog reductase after preincubation (3 min) with antimycin A (75 nM), which gave the same result as with cyanide. Complex III was assayed as reduced Q analog-cytochrome-*c* oxidoreductase in the presence of cyanide.

Inhibitor	Inhibition pattern obtained with varied Q analog substrate	
	DPB	DB
Complex I		
Compound 8	Competitive	Competitive
Compound 16	Competitive	Competitive
5′-Pentyl-MPP^+ + TPB^-	Mixed	ND
Complex II		
Compound 8	Noncompetitive	Noncompetitive
Compound 16	Noncompetitive	Noncompetitive
Complex III		
Antimycin A	Uncompetitive	Uncompetitive
Myxothiazol	Noncompetitive	Noncompetitive
Compound 8	Competitive	Uncompetitive
Compound 16	Competitive	Uncompetitive

which differs from compound 6 only in having a chloro group instead of the hydroxyl group at position 1. Compound 6 gives a maximum inhibition of only 55% (Table II). Thus, the phenolic OH group is not essential for the inhibition of Complex I activity, and indeed, the chloro group improves the inhibition.

Inhibition of Succinate-Coenzyme Q Oxidoreductase by 2-Substituted 4,6-Dinitrophenols—Previous studies with neurotoxins such as MPP^+ and β-carbolines showed that inhibition



SCHEME 1. **Proposed kinetic mechanism for Complex III.** *Cyt C*, cytochrome *c*.

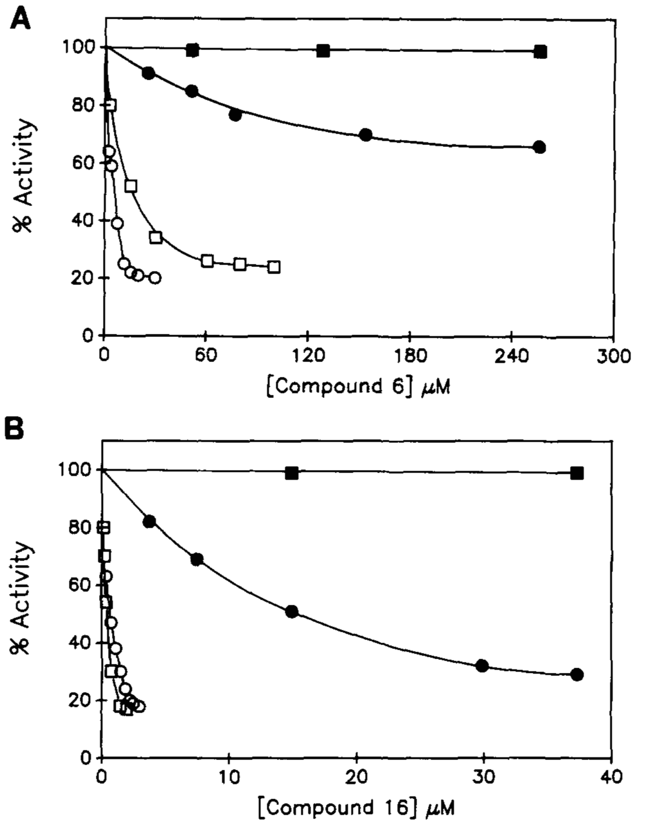


FIG. 5. **Typical inhibition curves for 4,6-dinitrophenol analogs on electron transport complexes.** *A*, compound 6; *B*, compound 16. The activities measured were as follows: Complex I (○), Complex II (●), Complex III (□), and Complex IV (■).

of Complex II activity is observed at concentrations 1–2 orders of magnitude higher compared with inhibition of Complex I (25, 26). We therefore examined the effect of compounds 1–3, 6, 8, 10, and 16 on Complex II (Table II). The dinitrophenol derivatives exhibited a poor inhibitory effect (all <45%), except for compounds 8 and 16, which inhibited Complexes I and II to almost the same extent. However, the I_H for compound 8 was 56 μM, and that for compound 16 was 9.5 μM, compared to 0.1 and 0.43 μM, respectively, for Complex I. Except for compound 6A, which inhibited Complex II by only 32%, the inhibitory potency, as measured by I_H, for all the other compounds was much higher (3–533-fold) for Complex I than for Complex II. In general, the more hydrophobic the 2-substituent, the better the inhibition of Complex II activity.

Inhibition of Complex III Activity—The inhibitory activity in terms of I_H with Complex III was determined for compounds 6, 8–10, 16, and 6A (Table II). The I_H values obtained for Complex III correspond to those for Complex I, indicating that the structural requirements for inhibitor binding are nearly the same for both. This suggests that the ubiquinone-binding sites in Complexes I and III may have similar structures. It is notable that although the I_H value for compound 6A with Complex III (19 μM) was almost identical to that with Complex I (23 μM), the maximum percent inhibition of Complex III by compound 6A was only 34%, which is much lower than that with Complex I (74%). Thus, the phenolic OH group is important for a strong inhibition of the catalytic reaction of Complex III.

DISCUSSION

This investigation was undertaken for two purposes. The first was to determine whether certain 2-substituted 4,6-dinitrophenols might not serve as universal inhibitors of respiratory chain complexes in which Q plays a functional role. As seen in Table II, compounds 8 and 16 seem to qualify as such, although even these inhibitors are far more effective in blocking electron transport in Complexes I and III than in Complex II. All the other compounds tested inhibited Complexes I and III, but not Complex II. Extension of these studies to other inhibitors of this type, designed on the basis of these data, may reveal common requirements for efficient binding to and inhibition of all three complexes.

The second purpose was to explore similarities and differences in the structural requirements for effective interaction of this class of inhibitors with the ubiquinone-binding sites in Complexes I, II, and III. Systematic structure-activity studies to complement information from site-directed mutagenesis can provide useful information on the structures of the substrate sites until such time that their tertiary structures are established.

Similar structure-inhibition patterns were obtained with Complexes I and III from beef heart mitochondria. These patterns were consistent with those observed for Complex III in rat liver mitochondria (17) and indicate that the structural requirements for the good fit of the inhibitor molecule in ubiquinone-binding sites may be similar in these complexes. Several trends can be identified. 1) The inhibitory activity tended to increase as the length of the main chain of the 2-substituent increased. 2) The existence of an α-branching structure in the 2-substituent was favorable for inhibition. 3) A main chain length of four carbons with an α-methyl branch (compound 8) gave the maximum potency. In general, the hydrophobicity of inhibitor molecules can play an important role in the inhibitory action on biological activities by enhancing membrane (lipid) solubility and hydrophobic interaction with the binding site. In this sense, the hydrophobic effect should be distinguished from other structural factors deciding inhibitor binding. The effect of an α-branching structure is apparent in Table II when one compares the inhibitory activities of inhibitors possessing the same hydrophobicity (compound 6 *versus* 10 and compound 16 *versus* 15 or 23). On the other hand, the significance of the hydrophobic effect is indicated by comparing the activities of compounds 6 and 8, both carrying an α-branching structure. Considering that the activity of the most hydrophobic inhibitor (23) that lacks an α-branching structure was much poorer than that of the inhibitors possessing an α-branching structure such as compounds 8, 9, and 16, it is likely that the effect of an α-branching structure on inhibitor

TABLE II
Structure of 2-substituted 4,6-dinitrophenols and their inhibition of NADH oxidase activity in beef heart submitochondrial particles

Compound	Substituent (2-position)	π^a	Complex I		Complex II		Complex III	
			I_H	Maximum inhibition	I_H	Maximum inhibition	I_H	Maximum inhibition
			μM	%	μM	%	μM	%
1	H	0	62	84				
2	CH_3	0.56	170	82	1450	26		
3	CH_2CH_3	1.02	164	48	410	18		
4	$CH(CH_3)_2$	1.53	7	62				
6	$CH(CH_3)CH_2CH_3$	1.96	3	55	54	37	11	74
6A[b]	$CH(CH_3)CH_2CH_3$	1.96	23	90	25	32	19	34
7	$CH(CH_3)_3$	1.98	67	63				
8	$CH(CH_3)CH_2CH_2CH_3$	2.54	0.10	73	56	65	0.13	88
9	$CH(CH_2CH_3)_2$	2.54	0.19	64			0.26	84
10	$CH_2CH(CH_3)_2$	1.96	85	64	260	43	68	38
15	$CH_2CH_2CH(CH_3)CH_2CH_3$	3.08	0.83	88				
16	$CH(CH_3)(CH_2)_3CH_3$	3.08	0.43	80	9.5	71	0.29	82
23	$CH_2(CH_2)_4CH_3$	3.21	38	73				

[a] The π value is the hydrophobicity index estimated from the partition coefficient in octanol/water.
[b] Compound 6A has a Cl group instead of an OH group.

binding is more important than that of hydrophobic interaction, providing that the inhibitor of interest has hydrophobicity of >1.5 or so. Saitoh *et al.* (17) studied the effect of an α-branching structure in detail; the α-branching structure makes the main chain extend almost perpendicular to the benzene ring plane because of structural congestion between the α-methyl and phenolic OH groups, resulting in a configuration similar to the portion of the isoprenoid side chain next to the natural ubiquinone ring. The correspondence in structural requirements for inhibitor binding between Complexes I and III suggests that the ubiquinone-binding sites share some common structural characteristics. Heinrich *et al.* (27) recently reported data that also suggest similarity of the structures of ubiquinone-binding sites in Complexes I and III from different energy-transducing biomembranes. They showed that the amino acid sequence for the 9.5-kDa ubiquinone-binding polypeptide in Complex I from *Neurospora crassa* shows a similarity to a putative ubiquinol-binding subunit (also a 9.5-kDa polypeptide (28)) in Complex III from bovine heart mitochondria.

In contrast, the inhibitory potency and the structural requirements for inhibitor binding with Complex II differed greatly from those with Complexes I and III, suggesting that the three-dimensional structure of the ubiquinone-binding site in the former is quite different. Electron paramagnetic resonance studies indicate that the pair of semiquinone radicals detected in Complex II lie perpendicular to the membrane (5) so that the greater hydrophobicity, which would carry the inhibitor further into the membrane, would bring it closer to the location of electron transfer. The α-branching structure is less favorable for Complex II inhibition than was found for Complex III inhibition (17) and for Complex I inhibition (Table II). The I_H values for compounds without α-branching (such as compounds 3 and 10) differ by only 3-fold between Complexes I and II, whereas the I_H values for those with an α-methyl group on the 2-substituent (such as compounds 6, 8, and 16) are >10-fold lower in Complex I than in Complex II.

During these studies, it became apparent that a third point can be made from these data. The inhibition patterns observed for Complex III suggest that there are two binding sites for the Q reductant in the Q_o region and that these sites distinguish between $DPBH_2$ and DBH_2. This is in accord with the double-occupancy Q_o site model proposed recently on the basis of electron paramagnetic resonance studies on the cytochrome bc_1 complex of *Rhodobacter capsulatus* (29). The two binding domains, one with a strong and the other with a weak interaction with ubiquinol, could permit a full turnover without exchange with the Q pool. Based on the differences between $DPBH_2$ and DBH_2 observed here, the use of a series of ubiquinol analogs should reveal more information about these sites and perhaps permit further elaboration of the double-site model.

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